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Method of the Year 2024: spatial proteomics

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Approaches for profiling the spatial proteome in tissues are the basis of atlas-scale projects that are delivering on their promise for understanding biological complexity in health and disease.

Humans by nature are inquisitive. We love to explore, and as we do, we make maps. Perhaps it is then not surprising that we find ourselves in an age of atlas-building as we explore the incredible complexity of biological systems with cutting-edge methods to better understand how cells, tissues, organs and organisms connect structure and function.

In the spirit of exploring and mapping biological complexity, we have chosen spatial proteomics as our Method of the Year for its critical role in revealing the organization of complex tissues. Spatial proteomics is an umbrella term that covers a broad swath of immunohistochemistry-based methods including, but not limited to, cyclic immunofluorescence (cycIF), co-detection by indexing (CODEX), iterative bleaching extends multiplexity (IBEX), multiplexed ion beam imaging (MIBI) and imaging mass cytometry (IMC). These approaches can be used to generate highly multiplexed images of specimens such as tissue and organ slices to understand their protein composition and spatial organization and are the basis of many global atlas projects. We are also excited about a newer technique known as deep visual proteomics (DVP), in which complex samples are laser dissected and individual dissociated cells are analyzed by mass spectrometry in such a way that their spatial context information is retained to create spatial protein maps. A major benefit of this technique is that it is not limited by the number of available antibodies and thus achieves substantially greater proteome coverage.

Spatial proteomics technologies such as immunofluorescence have been around longer than other spatial biology methods, such as spatially resolved transcriptomics, which was our Method of the Year in 2021, so why are we choosing it as Method of the Year now? For one, we were excited by the recent

development of DVP and other methods seeking new ways to explore the spatial proteome in greater depth and breadth. In addition, we were inspired by the current efforts of large consortia such as the Human BioMolecular Atlas Program (HuBMAP) and the Human Tumor Atlas Network (HTAN) not only to create large atlases of data for the scientific and medical communities, but also to develop tools to process, analyze, visualize and mine the data to go beyond the pretty pictures and deeper into biological discovery. This month's issue features two papers from the HTAN consortium: an Article from Peter Sorger and colleagues[1] describing CyLinter, an improved tool for quality control of highly multiplexed images, and an Article from Benjamin Raphael and colleagues[2] presenting CalicoST, an algorithm to simultaneously infer allele-specific copy number aberrations and reconstruct spatial tumor evolution from spatially resolved transcriptomics data. For this month's News Feature, journalist Vivien Marx asked some researchers about atlas-building and where they see things going next[3].

Our special issue features a series of Comments on the past, present and future of spatial proteomics. The first piece[4], from Bernd Bodenmiller, introduces why proteins are such interesting targets for biological investigations and offers a brief look back at how immunofluorescence has grown into the field of spatial proteomics. The piece further describes how generating large atlases will help reveal the intricacies of complex tissues and pave the way for precision medicine. It ends with a discussion of technological developments that will help move the field forward, briefly commenting on the role artificial intelligence may play in the future.

Computational tools for spatial proteomics are the focus of the second Comment, from Yuval Bussi and Leeat Keren[5]. These authors note that current image processing and analysis workflow are well defined but fragmented, with various steps happening back to back rather than in an integrated fashion. They envision a future for the field where image processing and analysis steps work in concert for improved biological discovery.

The third Comment[6], from Daniela Quail and Logan Walsh, discusses how spatial proteomics has revolutionized cancer research, from our understanding of tissue organization and cell–cell interactions to how it has shaped our thinking on how the immune system interacts with tumors. The piece also covers the potential role of spatial proteomics in combination with artificial intelligence in generating hypotheses for basic research, improving personalized medicine, and guiding future therapeutic strategies to fight cancer.

The fourth Comment, from Thierry Nordmann, Andreas Mund and Matthias Mann[7], introduces deep visual proteomics and the benefits of using mass spectrometry to probe the complexity of the proteome during processes such as development or in disease. They note that future improvements to sensitivity will allow mass spectrometry access to the entire proteome, including post-translational modifications, with single-cell resolution. They also discuss the benefits of combining DVP with other 'omics methods, future efforts to democratize the technology, and moving the technology into the clinic.

Is spatial proteomics enough? The final Comment, from Rong Fan[8], discusses spatial proteomics in the context of other 'omics technologies and the importance of integrating complementary technologies such as spatial transcriptomics and spatial epigenetic profiling to gain a more holistic understanding of biological complexity. This piece

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discusses the history of spatial 'omics, existing protein-inclusive multiomics methods, and the importance of next-generation computational tools for registering and integrating information from different modalities across volumes.

Although the present capabilities of spatial proteomics technologies are remarkable, we are also excited for the future of this field. We expect we will see the expanded application of these technologies toward growing existing and new atlas projects, with efforts being made concurrently to improve metadata handling, quality control and democratization of these methods. In addition, we think the application of multiomics methods that add the most complementary information to what is already present in spatial proteomics will continue to grow. We also expect that more is more, and that technological improvements in sample preparation (better affinity reagents, labeling, barcoding, signal amplification, sectioning) and microscopy (resolution, field of view) will lead us to spatial proteomics with higher coverage, over larger 3D volumes, and with subcellular resolution or even super-resolution, as discussed in our Methods to Watch on subcellular spatial proteomics[9] in this issue. With these data will come even more insight into how healthy cells work in their rightful context within tissues and what changes when things go wrong.

We are also excited for more than just spatial proteomics! We invite you to explore the rest of our special issue content, which includes eight Methods to Watch about which we are especially enthusiastic. We sincerely wish you and yours a safe and happy holiday season and a happy new year.

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